Chapter 13: Probability

School Worksheet: step-wise solutions

20 questions **51** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ — 1 mark

If $P(A) = 0.8$, $P(B) = 0.5$ and $P(B|A) = 0.4$, then $P(A \cap B)$ is:

(A) 0.32 **(B)** 0.25

(C) 0.1 **(D)** 0.4

Formula / idea: Multiplication rule: $P(A \cap B) = P(A)\,P(B|A)$.

1 Write the multiplication rule of probability.
$P(A \cap B) = P(A) \cdot P(B|A)$

2 Substitute the given values. ✓ 1
$P(A \cap B) = 0.8 \times 0.4 = 0.32$

3 Pick the matching option.
Option (A)

Answer **(A)** 0.32

Watch out: Using $P(B)$ instead of $P(A)$: the condition in $P(B|A)$ is A, so multiply by $P(A)$.

2 MCQ 1 mark

The probability that a particular item is available in three shops A, B and C is $\frac{4}{5}$, $\frac{3}{4}$ and $\frac{2}{3}$ respectively. If a person visits all three shops, the probability that the item is available in at least one of them is:

(A) $\frac{59}{60}$

(B) 1

(C) $\frac{1}{60}$

(D) $\frac{9}{60}$

Formula / idea: $P(\text{at least one}) = 1 - P(\text{none})$, for independent events.

1 Find the probability that the item is NOT available in each shop.

$P(A') = \frac{1}{5}, \quad P(B') = \frac{1}{4}, \quad P(C') = \frac{1}{3}$

2 Shops are independent, so multiply to get 'available in none'.

$P(\text{none}) = \frac{1}{5} \cdot \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{60}$

3 Subtract from 1. ✓ 1

$P(\text{at least one}) = 1 - \frac{1}{60} = \frac{59}{60}$

4 Pick the matching option.

Option (A)

Answer **(A)** $\frac{59}{60}$

Watch out: Multiplying the three given probabilities: that is 'available in all three', not 'at least one'.

3 MCQ 1 mark

A die is thrown. If E is the event "the number is 4" and F is the event "the number is even", then $P(E|F)$ is:

(A) $\frac{1}{6}$

(B) $\frac{1}{2}$

(C) $\frac{1}{3}$

(D) $\frac{2}{3}$

Formula / idea: $P(E|F) = \frac{P(E \cap F)}{P(F)}, P(F) \neq 0.$

1 Write the events.

$E = \{4\}, \quad F = \{2, 4, 6\}, \quad E \cap F = \{4\}$

2 Find the probabilities.

$P(F) = \frac{3}{6} = \frac{1}{2}, \quad P(E \cap F) = \frac{1}{6}$

3 Apply the conditional probability formula. ✓ 1

$P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{1/6}{1/2} = \frac{1}{3}$

4 Pick the matching option.

Option (C)

Answer **(C)** $\frac{1}{3}$

Watch out: Answering $\frac{1}{6}$: once F is given, only 3 outcomes remain, not 6.

4 MCQ 1 mark

If $P(A) = \frac{1}{2}$ and $P(B) = 0$, then $P(A|B)$ is:

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) **not defined**

Formula / idea: $P(A|B) = \frac{P(A \cap B)}{P(B)}$ is defined only when $P(B) \neq 0$.

1 Write the formula.
$P(A|B) = \frac{P(A \cap B)}{P(B)}$

2 Here the denominator is zero, so the ratio has no value. ✓ 1
$P(B) = 0 \Rightarrow P(A|B)$ is not defined

3 Pick the matching option.
Option (D)

Answer **(D) not defined**

Watch out: Choosing 0: the formula needs $P(B) \neq 0$, so no value can be given.

5 MCQ 1 mark

A random variable X has the distribution $P(X = 0) = k$, $P(X = 1) = 2k$, $P(X = 2) = 3k$. The value of k is:

(A) $\frac{1}{6}$

(B) $\frac{1}{3}$

(C) $\frac{1}{2}$

(D) 1

Formula / idea: In a probability distribution, $\sum P(X = x_i) = 1$.

1 Sum of all probabilities is 1.
$k + 2k + 3k = 1$

2 Solve for k. ✓ 1
$6k = 1 \Rightarrow k = \frac{1}{6}$

3	Pick the matching option. Option (A)	

Answer **(A)** $\frac{1}{6}$

6 Assertion-Reason — 1 mark

Assertion (A): Events A and B with $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{6}$ are independent.
Reason (R): Two events A and B are independent if and only if $P(A \cap B) = P(A)\,P(B)$.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: A, B independent $\iff P(A \cap B) = P(A)\,P(B)$.

1	Check A: find $P(A)\,P(B)$. $P(A)\,P(B) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6} = P(A \cap B)$	
2	So A and B are independent. **Assertion is true.**	
3	Reason is the definition of independent events. **Reason is true.**	
4	A is proved exactly by the test stated in R. **R explains A.**	
5	Pick the matching option. Option (A)	✓ 1

Answer **(A) Both A and R are true and R is the correct explanation of A**

7 Short answer — 2 marks

Two cards are drawn one after the other without replacement from a well-shuffled pack of 52 cards. Find the probability that both are kings.

Formula / idea: Multiplication theorem: $P(E \cap F) = P(E)\,P(F|E)$.

1	Let E: first card is a king, F: second card is a king.	
2	Write the multiplication theorem. $P(E \cap F) = P(E) \cdot P(F|E)$	✓ ½

3 4 kings in 52 cards. ✓ ½

$$P(E) = \frac{4}{52}$$

4 One king removed: 3 kings left in 51 cards. ✓ ½

$$P(F|E) = \frac{3}{51}$$

5 Multiply and simplify. ✓ ½

$$P(E \cap F) = \frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}$$

6 Required probability $= \frac{1}{221}$.

Answer $\frac{1}{221}$

Watch out: Using $\frac{4}{52} \times \frac{4}{52}$: without replacement, the second draw has 3 kings in 51 cards.

Level 2

18 marks

8 **Short answer** 3 marks

The chances of P, Q and R getting selected as CEO of a company are in the ratio $4 : 1 : 2$. The probabilities that the company increases its profits under the new CEO P, Q or R are 0.3, 0.8 and 0.5 respectively. If the company increased its profits, find the probability that it is due to the appointment of R as CEO.

Formula / idea: Bayes' theorem: $P(E_i|A) = \dfrac{P(E_i)P(A|E_i)}{\sum P(E_j)P(A|E_j)}$.

1 Let E_1, E_2, E_3: P, Q, R is selected. Let A: profits increase.

2 Ratio $4 : 1 : 2$, total 7 parts. ✓ 1

$$P(E_1) = \frac{4}{7}, \quad P(E_2) = \frac{1}{7}, \quad P(E_3) = \frac{2}{7}$$

3 Write the given conditional probabilities. ✓ ½

$$P(A|E_1) = 0.3, \quad P(A|E_2) = 0.8, \quad P(A|E_3) = 0.5$$

4 Find $P(A)$ by the theorem of total probability. ✓ ½

$$P(A) = \frac{4}{7}(0.3) + \frac{1}{7}(0.8) + \frac{2}{7}(0.5)$$
$$= \frac{1.2 + 0.8 + 1.0}{7} = \frac{3}{7}$$

5 Apply Bayes' theorem for E_3. ✓ ½

$$P(E_3|A) = \frac{P(E_3)\,P(A|E_3)}{P(A)} = \frac{\frac{2}{7}(0.5)}{\frac{3}{7}}$$

6 Simplify.
$P(E_3|A) = \dfrac{1/7}{3/7} = \dfrac{1}{3}$ ✓ ½

7 Required probability $= \dfrac{1}{3}$.

Answer $P(E_3|A) = \dfrac{1}{3}$

Watch out: Answering $P(A|E_3) = 0.5$: the question asks for $P(E_3|A)$, the reverse direction.

9 Short answer — 2 marks

Find the probability distribution of the number of heads in two tosses of a fair coin. Also find its mean.

Formula / idea: Mean $E(X) = \sum x_i\, p_i$.

1 Sample space and values of X (number of heads). ✓ ½
$S = \{HH, HT, TH, TT\}, \quad X = 0, 1, 2$

2 Find each probability.
$P(X=0) = P(TT) = \frac{1}{4},\ P(X=1) = P(HT, TH) = \frac{2}{4} = \frac{1}{2},\ P(X=2) = P(HH) = \frac{1}{4}$

3 Write the probability distribution. ✓ 1

X	0	1	2
$P(X)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

4 Find the mean. ✓ ½
$E(X) = 0 \cdot \dfrac{1}{4} + 1 \cdot \dfrac{1}{2} + 2 \cdot \dfrac{1}{4} = 0 + \dfrac{1}{2} + \dfrac{1}{2} = 1$

5 Mean number of heads $= 1$.

Answer X**:** $0, 1, 2$ **with** $P(X)$**:** $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$**; mean** $= 1$

Watch out: Taking $P(X=1) = \dfrac{1}{4}$: both HT and TH give one head.

10 Short answer — 3 marks

Given $P(A) = 0.6$, $P(B) = 0.3$ and $P(A \cap B) = 0.2$, find $P(A|B)$, $P(B|A)$ and $P(A'|B')$.

Formula / idea: $P(A|B) = \dfrac{P(A \cap B)}{P(B)}$ and $A' \cap B' = (A \cup B)'$.

1 Find $P(A|B)$. ✓ 1
$P(A|B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{0.2}{0.3} = \dfrac{2}{3}$

2 Find $P(B|A)$.
$P(B|A) = \dfrac{P(A \cap B)}{P(A)} = \dfrac{0.2}{0.6} = \dfrac{1}{3}$ ✓ 1

3 Find $P(A \cup B)$ by the addition rule.
$P(A \cup B) = 0.6 + 0.3 - 0.2 = 0.7$

4 Use De Morgan's law for $P(A' \cap B')$.
$P(A' \cap B') = P\big((A \cup B)'\big) = 1 - 0.7 = 0.3$ ✓ ½

5 Find $P(B')$.
$P(B') = 1 - 0.3 = 0.7$

6 Find $P(A'|B')$.
$P(A'|B') = \dfrac{P(A' \cap B')}{P(B')} = \dfrac{0.3}{0.7} = \dfrac{3}{7}$ ✓ ½

7 Hence $P(A|B) = \dfrac{2}{3}$, $P(B|A) = \dfrac{1}{3}$, $P(A'|B') = \dfrac{3}{7}$.

Answer $P(A|B) = \dfrac{2}{3}$, $P(B|A) = \dfrac{1}{3}$, $P(A'|B') = \dfrac{3}{7}$

Watch out: Writing $P(A' \cap B') = P(A')P(B')$: that needs independence, which is not given.

11 Short answer — 3 marks

A and B independently try to solve a problem with probabilities $\dfrac{1}{2}$ and $\dfrac{1}{3}$ respectively. Find the probability that (a) the problem is solved (b) exactly one of them solves it.

Formula / idea: For independent events, $P(A \cap B) = P(A)P(B)$; also A', B' are independent.

1 Write the given and complementary probabilities.
$P(A) = \dfrac{1}{2},\ P(B) = \dfrac{1}{3},\ P(A') = \dfrac{1}{2},\ P(B') = \dfrac{2}{3}$ ✓ ½

2 (a) Problem is solved means at least one solves it.
$P(\text{solved}) = 1 - P(A' \cap B') = 1 - P(A')\,P(B')$ ✓ ½

3 Substitute and simplify.
$P(\text{solved}) = 1 - \dfrac{1}{2} \cdot \dfrac{2}{3} = 1 - \dfrac{1}{3} = \dfrac{2}{3}$ ✓ ½

4 (b) Exactly one solves: A solves and B fails, or A fails and B solves.
$P = P(A)\,P(B') + P(A')\,P(B)$ ✓ ½

5 Substitute.
$P = \dfrac{1}{2} \cdot \dfrac{2}{3} + \dfrac{1}{2} \cdot \dfrac{1}{3} = \dfrac{1}{3} + \dfrac{1}{6}$ ✓ ½

6 Simplify.
$P = \dfrac{2+1}{6} = \dfrac{1}{2}$ ✓ ½

7 Hence (a) $\dfrac{2}{3}$, (b) $\dfrac{1}{2}$.

Answer **(a)** $\dfrac{2}{3}$ **(b)** $\dfrac{1}{2}$

Watch out: For (a), adding $\frac{1}{2} + \frac{1}{3}$: this counts 'both solve' twice.

12 Short answer — 2 marks

A family has two children. Find the probability that both are boys, given that at least one of them is a boy.

Formula / idea: $P(E|F) = \dfrac{P(E \cap F)}{P(F)}$.

1 Write the sample space (b = boy, g = girl).
$S = \{bb, bg, gb, gg\}$ ✓ ½

2 Let E: both are boys, F: at least one is a boy.
$E = \{bb\}, \quad F = \{bb, bg, gb\}, \quad E \cap F = \{bb\}$ ✓ ½

3 Find the probabilities.
$P(F) = \dfrac{3}{4}, \quad P(E \cap F) = \dfrac{1}{4}$

4 Apply the conditional probability formula.
$P(E|F) = \dfrac{P(E \cap F)}{P(F)} = \dfrac{1/4}{3/4} = \dfrac{1}{3}$ ✓ 1

5 Required probability $= \dfrac{1}{3}$.

Answer $\dfrac{1}{3}$

Watch out: Answering $\frac{1}{2}$: bg and gb are two different outcomes, so F has 3 outcomes.

13 Short answer — 3 marks

A random variable X has $P(X = x) = kx$ for $x = 1, 2, 3, 4$ and 0 otherwise. Find k, $P(X \geq 3)$ and the mean of X.

Formula / idea: $\sum p_i = 1$ and mean $E(X) = \sum x_i p_i$.

1 Write the distribution.

X	1	2	3	4
$P(X)$	k	$2k$	$3k$	$4k$

2 Sum of probabilities is 1. ✓ 1

$$k + 2k + 3k + 4k = 1 \;\Rightarrow\; 10k = 1 \;\Rightarrow\; k = \frac{1}{10}$$

3 Find $P(X \geq 3)$. ✓ 1

$$P(X \geq 3) = P(X = 3) + P(X = 4) = 3k + 4k = 7k = \frac{7}{10}$$

4 Find the mean. ✓ ½

$$E(X) = \sum x \cdot kx = k(1 + 4 + 9 + 16) = \frac{30}{10}$$

5 Simplify. ✓ ½

$$E(X) = 3$$

6 Hence $k = \frac{1}{10}$, $P(X \geq 3) = \frac{7}{10}$, mean $= 3$.

Answer $k = \frac{1}{10}$, $P(X \geq 3) = \frac{7}{10}$, **mean** $= 3$

Watch out: Using $\sum kx$ instead of $\sum x \cdot kx$ for the mean: each value must be multiplied by its probability.

14 Short answer — 2 marks

Three coins are tossed. Let E: "three heads or three tails" and F: "at least two heads". Are E and F independent?

Formula / idea: E, F independent $\iff P(E \cap F) = P(E)\,P(F)$.

1 Write the sample space (8 outcomes).

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

2 Write E and find $P(E)$. ✓ ½

$$E = \{HHH, TTT\}, \quad P(E) = \frac{2}{8} = \frac{1}{4}$$

3 Write F and find $P(F)$. ✓ ½

$$F = \{HHH, HHT, HTH, THH\}, \quad P(F) = \frac{4}{8} = \frac{1}{2}$$

4 Find $E \cap F$. ✓ ½

$$E \cap F = \{HHH\}, \quad P(E \cap F) = \frac{1}{8}$$

5 Compare with $P(E)\,P(F)$.

$$P(E)\,P(F) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8} = P(E \cap F)$$

6	Hence E and F are independent.	✓ ½

Answer **Yes, $P(E \cap F) = \frac{1}{8} = P(E)\,P(F)$, so E and F are independent**

Watch out: Forgetting HHH in F: "at least two heads" includes three heads.

Level 3

25 marks

15 **Long answer** 5 marks

A factory has machines A, B and C producing 50%, 30% and 20% of the items, of which 2%, 3% and 4% respectively are defective. An item picked at random is found defective. Find the probability that it was produced by machine A.

Formula / idea: Bayes' theorem: $P(E_i|D) = \dfrac{P(E_i)P(D|E_i)}{\sum P(E_j)P(D|E_j)}$.

1	Let E_1, E_2, E_3: item made by A, B, C. Let D: item is defective.	✓ 1
2	Write the prior probabilities. $P(E_1) = 0.5, \quad P(E_2) = 0.3, \quad P(E_3) = 0.2$	✓ 1
3	Write the probabilities of a defective item from each machine. $P(D\|E_1) = 0.02, \quad P(D\|E_2) = 0.03, \quad P(D\|E_3) = 0.04$	✓ 1
4	State Bayes' theorem for E_1. $P(E_1\|D) = \dfrac{P(E_1)P(D\|E_1)}{P(E_1)P(D\|E_1) + P(E_2)P(D\|E_2) + P(E_3)P(D\|E_3)}$	
5	Substitute the values. $P(E_1\|D) = \dfrac{0.5 \times 0.02}{0.5 \times 0.02 + 0.3 \times 0.03 + 0.2 \times 0.04}$	✓ 1
6	Simplify. $P(E_1\|D) = \dfrac{0.010}{0.010 + 0.009 + 0.008} = \dfrac{0.010}{0.027} = \dfrac{10}{27}$	✓ 1
7	Required probability $= \dfrac{10}{27}$.	

Answer $P(E_1|D) = \dfrac{10}{27}$

Watch out: Writing $P(E_1|D) = 0.02$: that is $P(D|E_1)$, the reverse conditional.

16 **Short answer** 3 marks

In answering a multiple-choice question (4 options), a student either knows the answer or guesses. The probability that she knows the answer is $\frac{3}{4}$. If she answers correctly, find the probability that she knew the answer.

Formula / idea: Bayes' theorem with two cases: knows (E_1) or guesses (E_2).

1 Let E_1: she knows the answer, E_2: she guesses, A: answer is correct.

2 Write the prior probabilities. ✓ 1

$$P(E_1) = \frac{3}{4}, \quad P(E_2) = 1 - \frac{3}{4} = \frac{1}{4}$$

3 If she knows, she is surely correct; a guess is right 1 time in 4. ✓ 1

$$P(A|E_1) = 1, \quad P(A|E_2) = \frac{1}{4}$$

4 Apply Bayes' theorem. ✓ ½

$$P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)} = \frac{\frac{3}{4} \cdot 1}{\frac{3}{4} \cdot 1 + \frac{1}{4} \cdot \frac{1}{4}}$$

5 Simplify. ✓ ½

$$P(E_1|A) = \frac{\frac{12}{16}}{\frac{13}{16}} = \frac{12}{13}$$

6 Required probability $= \frac{12}{13}$.

Answer $P(E_1|A) = \frac{12}{13}$

Watch out: Taking $P(A|E_1) = \frac{3}{4}$: if she knows the answer she is correct with probability 1.

17 **Long answer** 5 marks

Two cards are drawn successively without replacement from a well-shuffled pack of 52 cards. Find the probability distribution of the number of aces and its mean.

Formula / idea: Without replacement: $P(E \cap F) = P(E)\,P(F|E)$; mean $= \sum x_i p_i$.

1 Let X = number of aces. 4 aces, 48 non-aces. X takes values $0, 1, 2$. ✓ ½

2 $X = 0$: both cards non-aces. ✓ 1

$$P(X = 0) = \frac{48}{52} \times \frac{47}{51} = \frac{2256}{2652} = \frac{188}{221}$$

3 $X = 1$: ace then non-ace, or non-ace then ace. ✓ 1

$$P(X = 1) = \frac{4}{52} \times \frac{48}{51} + \frac{48}{52} \times \frac{4}{51} = \frac{384}{2652} = \frac{32}{221}$$

4 $X = 2$: both cards aces. ✓ ½

$$P(X=2) = \frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}$$

5 Write the probability distribution. ✓ ½

X	0	1	2
$P(X)$	$\frac{188}{221}$	$\frac{32}{221}$	$\frac{1}{221}$

6 Check: probabilities add to 1.

$$\frac{188+32+1}{221} = \frac{221}{221} = 1$$

7 Write the mean formula and substitute. ✓ ½

$$E(X) = \sum x_i p_i = 0 \cdot \frac{188}{221} + 1 \cdot \frac{32}{221} + 2 \cdot \frac{1}{221}$$

8 Simplify. ✓ 1

$$E(X) = \frac{32+2}{221} = \frac{34}{221} = \frac{2}{13}$$

9 Mean number of aces $= \frac{2}{13}$.

Answer X**:** $0, 1, 2$ **with** $P(X)$**:** $\frac{188}{221}, \frac{32}{221}, \frac{1}{221}$**; mean** $= \frac{2}{13}$

Watch out: Counting only ace-then-non-ace for $X = 1$: the other order must be added too.

18 Short answer — 3 marks

A random variable X has the distribution: X: $0, 1, 2, 3, 4, 5, 6, 7$ with $P(X)$: $0,\ k,\ 2k,\ 2k,\ 3k,\ k^2,\ 2k^2,\ 7k^2 + k$. Find k, $P(X < 3)$ and $P(X > 6)$.

Formula / idea: $\sum p_i = 1$, and every $p_i \geq 0$.

1 Sum of all probabilities is 1.

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

2 Collect terms to get a quadratic. ✓ ½

$$10k^2 + 9k - 1 = 0$$

3 Factorise.

$$(10k-1)(k+1) = 0 \ \Rightarrow\ k = \frac{1}{10} \text{ or } k = -1$$

4 Reject $k = -1$, since a probability cannot be negative. ✓ ½

$$k = \frac{1}{10}$$

5 Find $P(X < 3)$. ✓ 1

$$P(X<3) = P(0) + P(1) + P(2) = 0 + k + 2k = 3k = \frac{3}{10}$$

6 Find $P(X > 6)$. ✓ 1

$$P(X > 6) = P(X = 7) = 7k^2 + k = \frac{7}{100} + \frac{1}{10} = \frac{17}{100}$$

7 Hence $k = \frac{1}{10}$, $P(X < 3) = \frac{3}{10}$, $P(X > 6) = \frac{17}{100}$.

Answer $k = \frac{1}{10}$, $P(X < 3) = \frac{3}{10}$, $P(X > 6) = \frac{17}{100}$

Watch out: Keeping $k = -1$: then $P(X = 1) = -1$, which is impossible.

19 Long answer — 5 marks

A laboratory blood test is 99% effective in detecting a disease when it is present, but gives a false positive for 0.5% of healthy persons. If 0.1% of the population has the disease, find the probability that a person who tests positive actually has the disease.

Formula / idea: Bayes' theorem: $P(E_1|A) = \dfrac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)}$.

1 Let E_1: person has the disease, E_2: person is healthy, A: test is positive. ✓ 1

2 Write the prior probabilities. ✓ 1

$$P(E_1) = 0.1\% = 0.001, \quad P(E_2) = 1 - 0.001 = 0.999$$

3 Write the probabilities of a positive test. ✓ 1

$$P(A|E_1) = 99\% = 0.99, \quad P(A|E_2) = 0.5\% = 0.005$$

4 Apply Bayes' theorem.

$$P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)}$$

5 Substitute the values. ✓ 1

$$P(E_1|A) = \frac{0.001 \times 0.99}{0.001 \times 0.99 + 0.999 \times 0.005}$$

6 Multiply numerator and denominator by 10^6.

$$P(E_1|A) = \frac{990}{990 + 4995} = \frac{990}{5985}$$

7 Divide top and bottom by 45. ✓ 1

$$P(E_1|A) = \frac{22}{133}$$

8 Required probability $= \frac{22}{133}$ (about 0.165).

Answer $P(E_1|A) = \frac{22}{133}$

Watch out: Taking $P(A|E_2) = 0.995$: the false positive rate 0.5% is itself $P(A|E_2)$.

20 **Case study** 4 marks

Case Study: An NGO organises a charity event to distribute woollen caps to children in winter. The caps are in three boxes: Box I has 30 red caps, Box II has 20 red and 10 green caps, and Box III has 30 green caps. The probability that Box i is selected and a cap picked out is $\frac{i}{6}$, $i = 1, 2, 3$. A person selects a cap.

(i) What is the probability that he selects a red cap? 2 marks

(ii) If he selects a green cap, what is the probability that it has come from Box II? 2 marks

Formula / idea: Total probability for (i); Bayes' theorem for (ii).

(i) 2 marks

1 Let E_1, E_2, E_3: Box I, II, III is selected. Let R: red cap.

2 Write $P(E_i) = \frac{i}{6}$. ✓ ½
$P(E_1) = \frac{1}{6}, \quad P(E_2) = \frac{2}{6}, \quad P(E_3) = \frac{3}{6}$

3 Probability of red from each box. ✓ ½
$P(R|E_1) = \frac{30}{30} = 1, \; P(R|E_2) = \frac{20}{30} = \frac{2}{3}, \; P(R|E_3) = 0$

4 Theorem of total probability. ✓ ½
$P(R) = \frac{1}{6}(1) + \frac{2}{6}\left(\frac{2}{3}\right) + \frac{3}{6}(0) = \frac{3}{18} + \frac{4}{18}$

5 Simplify. ✓ ½
$P(R) = \frac{7}{18}$

6 Probability of a red cap $= \frac{7}{18}$.

Answer (i) $\frac{7}{18}$

(ii) 2 marks

1 Let G: green cap. Probability of green from each box. ✓ ½
$P(G|E_1) = 0, \quad P(G|E_2) = \frac{10}{30} = \frac{1}{3}, \quad P(G|E_3) = 1$

2 Find $P(G)$ by total probability. ✓ ½
$P(G) = \frac{1}{6}(0) + \frac{2}{6}\left(\frac{1}{3}\right) + \frac{3}{6}(1) = \frac{2}{18} + \frac{9}{18} = \frac{11}{18}$

3 Apply Bayes' theorem for Box II. ✓ ½
$P(E_2|G) = \frac{P(E_2)P(G|E_2)}{P(G)} = \frac{\frac{2}{6} \cdot \frac{1}{3}}{\frac{11}{18}}$

4 Simplify.

$$P(E_2|G) = \frac{2/18}{11/18} = \frac{2}{11}$$

✓ ½

5 Required probability $= \frac{2}{11}$.

Answer (ii) $\frac{2}{11}$

Watch out: Taking each box as equally likely ($\frac{1}{3}$): here $P(E_i) = \frac{i}{6}$.